

A Deep Learning Approach to Predict Malnutrition Status of 0-59 Month's Older Children in Bangladesh

Md Mehrab Shahriar, Mirza Shaheen Iqbal, Samrat Mitra, Amit Kumar Das

Department of Computer Science and Engineering

East West University

Dhaka, Bangladesh

E-mail: *mshahriar_ewu@outlook.com, iqbal.ewu@gmail.com, somrat_dk@yahoo.com, amit.csedu@gmail.com*

Abstract— The state of malnutrition can be considered as a predominant issue for a developing nation like Bangladesh. Since today's children are the future's workforce, it explicitly impacts to the economic improvement of Bangladesh. So, prevention of child malnutrition is the most foremost investigation at this stage. The study aims to classify malnutrition based on deep learning approach of predictive modeling on significant malnutrition features to predict malnutrition status of a 0-59 months' older child. To do so an Artificial Neural Network (ANN) approach is applied to Bangladesh Demographic and Health Survey 2014 (BDHS) children data. This study clarifies how a predictive model classifies the malnutrition condition. ANN approach shows the best accuracy with wasting, underweight, and stunting. In conclusion, determining the malnutrition status using deep learning approach is the most scientific way to deal with it both for policymakers and clinicians.

Keywords— *Malnutrition, BDHS, Tensorflow, Keras, ANN, Deep Learning*

I. INTRODUCTION

Nutrition is the essential support of health and human body development across the whole life cycle. The lack or excess of consumed calories is called malnutrition. Malnutrition is experienced mainly for a lack of essential vitamins and minerals or by obsessing over calories which refer to overweight resulting lot of serious health problems and diseases such as heart disease, diabetics, and stroke [2, 3, 7].

The consequence of malnutrition, malnourished people, tend falling ill, increase the risk of being infected and being fractured.

Bangladesh is one of the developing countries and child malnutrition is one of the biggest problem here. Almost, 3.5 million of all the deaths and one-third of morbidities and mortalities among children from newborn to 59 months older is caused by malnutrition globally [5, 6, 13, 8]. Currently, Bangladesh has approximately 162.9 million people, among

them nearly 15.2 million are children from newborn to 59 months older. It has been reported in [4] that 9.5million and more are stunted and 56% are underweight children and around 17% of children are wasted. The study of malnutrition on under-five children is a very demandable choice for the researcher for several decades. Previously, almost all researcher studied on various factors which are responsible for having malnutrition on child under-five [3, 17]. Additionally, all studies [2, 3, 17] determined the risk factors of malnutrition based on the deterministic model, but none of the studies explained or explored predictive modeling based on the topmost risk factors and determined the malnutrition status. Therefore, the goal of the study is to classify the malnutrition status of birth to 59 months' older child based on deep learning approach of predictive modeling on significant malnutrition features. It is believed that predictive modeling approach will add significant contribution to the nutrition research.

II. RELATED WORK

Under-five children are more inclined to malnutrition deficiencies than any other children. A malnourished baby can be defined as a clinical indication of nutrient inadequacy which indicates as underweight, wasting and stunting [14]. In whatever way, anthropometric indicators are often more useful for cost-effectiveness and availability. Several statistical methods have inspected to find out the responsible risk factors for malnutrition. Among the methods, Linear Regression and Logistic Regression are widely studied for a finding of being malnourished among 0 to 59 month's old child [13, 16, 10]. There are several models to explain nutrition measurement level for a child are rule induction classifier with receiver operating curve (ROC), Naïve Bayes [19], decision tree [18] and the association rules.

Very few studies used variable analysis and least squares approach to finding out the relation between the selected variables respect to malnutrition [15, 20, 9]. Some of the features have been extracted to find out the most significant

factors which are directly responsible for malnutrition. Most of them used anthropometric analysis based on WHO's Z-score model [1]. The characterizing parameters, Weight-for-Age Z-score (WHZ) indicates for being wasting, Height-for-Age Z-score (HAZ) for being stunting, Weight-for-Age Z-score (WAZ) for being underweight are used to define a malnourished child. Among these, WHZ is mostly considered for defining a malnourished child. In very few researches, HAZ, WHZ, WAZ has taken as the indicator for malnourished [16, 17, 11, 12].

Using the above-mentioned indices, statistical models are used to find out whether these risk factors actually have direct impacts on malnutrition. So, it would be more interesting to get to know how well a deep learning approach is used to train the machine if the new data come near future and how correctly it predicts malnourished child concerning fit model trained by the old dataset.

III. METHODOLOGY

In the next section, a brief description of the dataset has been provided. In subsequent sections, data preprocessing and proposed methodologies on preprocessed data using various software and language will be explained.

A. Data Source

The raw data has been taken from the Bangladesh Demographic and Health Survey 2014 (BDHS). It has been designed to cover both the urban and rural area under every seven divisions in Bangladesh. Among urban and rural area, the BDHS dataset consisted of different types of demographic e.g. birth record, household information record across all the seven divisions in Bangladesh. In birth record which consists of 6995 instances together with 1742 features (factors) which are used in this study. To study the malnutrition, it was very important to calculate the anthropometric measurement of every under-five child individually from those families. As a result, 6,995 children of under-five gave their height, weight and age information which have been used to calculate the anthropometric measurement.

B. Anthropometric Analysis

HAZ, WAZ, WHZ are widely used to examine the magnitude of malnutrition as stunting, underweight and wasting respectively. These provide various information about the growth of a child and body compositions to evaluate the nutrition level of any individuals. This nutrition level is classified on the basis of Z-scores. Three anthropometric z-scores indicating WAZ, HAZ, WHZ were calculated based on the 2006 Child Growth Standard defined by WHO. The z-scores calculate of an individual's value with respect to the distribution of the standard reference population which was calculated using macro provided by WHO. The mathematical formula of z-scores is defined as follows:

$$Z - score = \frac{W - M}{R} \quad (1)$$

$$WHZ = \frac{X - M}{R} \quad (2)$$

$$WAZ = \frac{Y - M}{R} \quad (3)$$

$$HAZ = \frac{Z - M}{R} \quad (4)$$

Here, W, X, Y, Z, M, R refer to an individual value (height or weight), children's weight-for-height, children's weight-for-age, children's height-for-age, a median of the WHO's reference population, and standard deviation of WHO's reference population respectively. If any of the z scores is less than -2.0 according to above equations, then it indicates malnutrition.

C. Feature Extraction

It was very challenging to define malnutrition for the under-five child using the secondary sources of data. Some of the features do not have any impact on malnutrition, in general. Therefore, the labeling of data and features extraction was very important. Using anthropometric measurement, the data has been labeled. In this research it has been found 16 most significant features by literature review. Those features were extracted with respect to Primary Sampling Unit (PSU). PSU is used to select individual child information from BDHS data source. Among 1,742 features in BDHS 2014, 16 most significant features were extracted.

D. Data Preprocessing

Coming from section D, we have designed two phases for better understanding which approach is going to be visited.

In Phase-I, the dataset was not still fit for learning our model because of some noisy and string data were present. So, data had been cleaned using mean imputation and changed to numeric values from string values using python's most usages '*numpy*' library.

In Phase-II, one-hot encoding is used to convert the categorical data to a better form which is understandable by deep learning algorithms for better prediction. To keep the data in a standard form, mean-normalization of feature scaling is calculated.

E. Proposed Model

From phase-II, the preprocessed data is now prepared to fit for our deep learning model.

Tensorflow, (numeric tool in Python) has been implemented on our deep learning model. *Tensorflow* uses data flow graphs to build models. It is really essential to building large-scale layers in an Artificial Neural Network (ANN) to analyze our model. '*Keras*', another popular library is used to build our model which runs *TensorFlow* at its backend to train the factors.

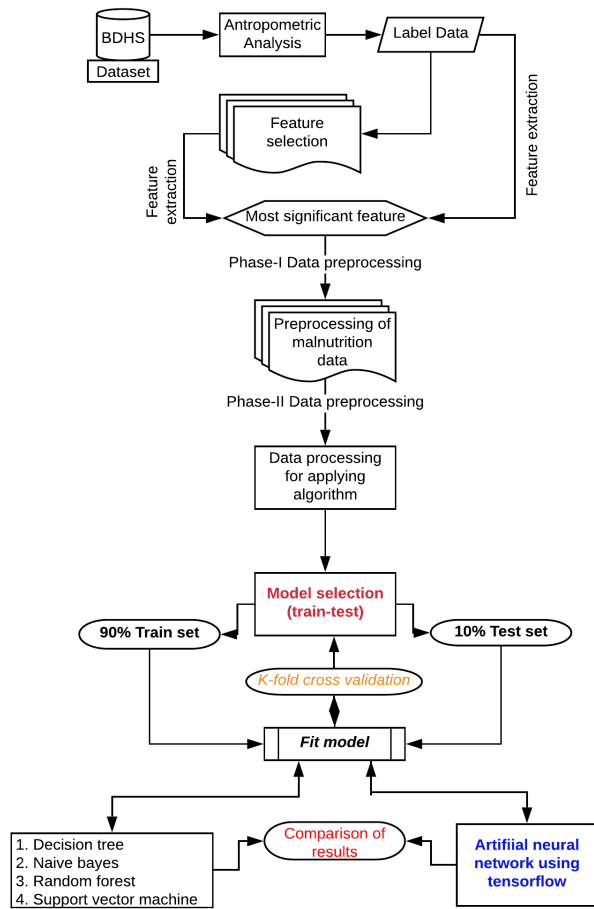


Figure 1. Model Framework.

After initializing the neural network, the model took 16 neurons as features in the input layer and 8 in the hidden layer. In this proposed model, ANN used three hidden layers after testing gradually one by one.

As this study purpose is to classify malnourished child from trained data, it sets a range (0, 1) of a linear function in ANN using the rectifier activation function is applied in hidden layers and sigmoid activation function in the output layer. This will give the probability of being malnourished as 1 and nourished as 0. As 90% of the data has been taken as training data and fit a model which runs 100-500 epochs.

Here each epoch is considered as one forward and one backward propagation. Finally, the most potent stochastic gradient descent optimizer parameter “rmsprop” is used. But it also experiments with “adam” another optimizer parameter of gradient descent which a perfect stochastic gradient descent algorithm is also. But “rmsprop” can find the best local optimal. The batch size is taken ten which means ten instances in each epoch at a time. Now in each epoch, it occurs some loss.

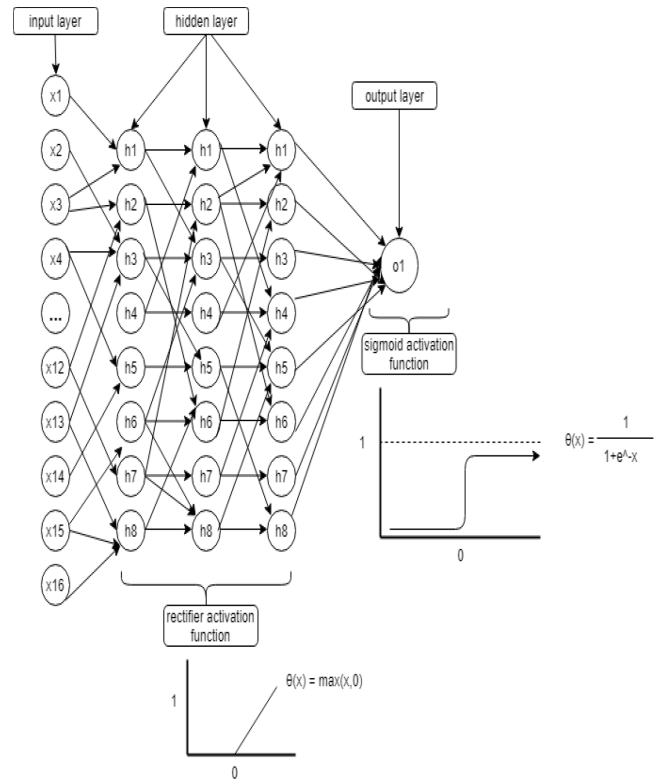


Figure 2. Artificial Neural Network Framework Based on Tensorflow.

To determine the losses, ‘*loss(binary_crossentropy)*’ function is used for classification. After calculating the loss, the best result is provided with ANN.

IV. RESULT ANALYSIS AND COMPARISON

The purpose of the study is to determine for being wasting, underweight and stunting child from the trained data, ANN approach shows the best result with accuracy close to 86.0%, 70.0%, and 67.30% respectively with wasting, underweight and stunting. These tests have been done with 10% of the data with k-fold(k=10)

cross-validation. Figure 3 shows the result of the accuracy of this approach. To calculate accuracy, we have used the formula shown below:

$$Accuracy = \frac{(TP + TN)}{(TP + FP + TP + FN)} \quad (6)$$

We have mainly implemented ANN on our dataset whereas another four machine learning algorithms have been used sequentially. We have also applied Support Vector Machine (SVM) classifier, Decision Tree classifier, Naive Bayes classifier, and Random Forest classifier besides ANN. Figure 4 shows the comparison among those four algorithms concerning ANN.

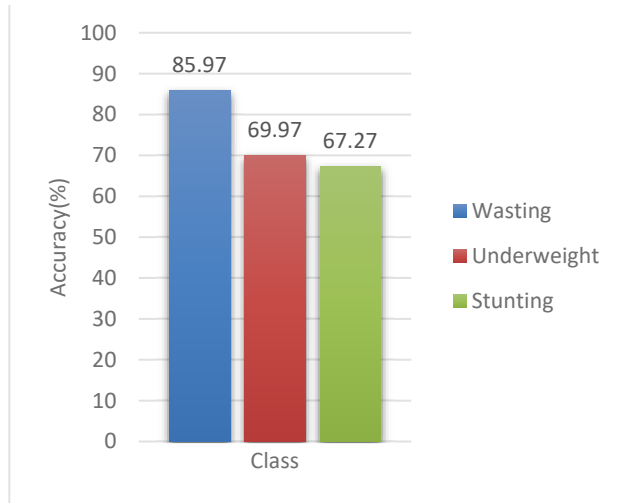


Figure 3. The accuracy of Wasting, Underweight, and Stunting.

The SVM gives also the closest result as compared to ANN with accuracy 84.93%, 67.75%, 63.34% for wasting, underweight and stunting respectively. But the value of precision, recall, and F1-score gives 0 by SVM and Random Forest, which means SVM and Random Forest failed to predict nourished child. But ANN gives much more good result than SVM and Random Forest.

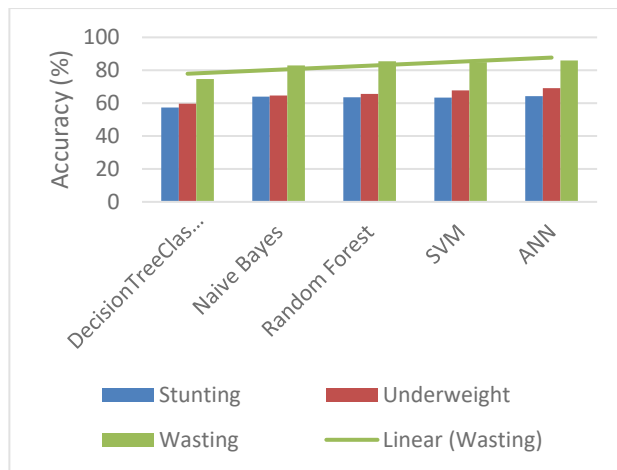


Figure 4. Accuracy Comparisons of All Algorithms.

In addition, the significant features have been analyzed again using backward elimination method on our predictive model. The features which are $p < 0.05$ (probabilistic value) considering statistically significant features and have a great impact on malnutrition otherwise the features are not a significant impact on malnutrition. We have also found out the non-significant features for our three-decision class. We have found some common features which do not have any effect on the dataset. Figure 5 shows that there has almost no impact on the father's education, mother's working (if mother

works outside the house) and mother's BMI of being wasting while mother's education has a tiny impact on wasting, stunting, and underweight. Although it is less significant overall as the probabilistic value is over 5% in all class. To distinguish a malnourished child from our trained data, the deep learning approach using Tensorflow performs better and gives a precise result from our predictive model with the highest accuracy.

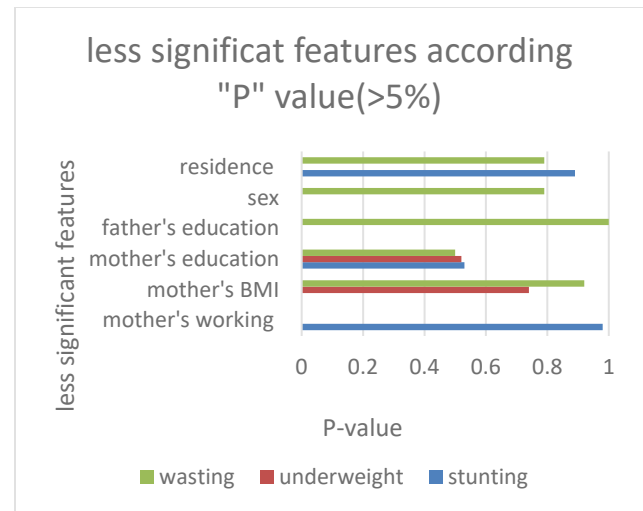


Figure 5. Less Significant Features

V. CONCLUSION

The human brainpower may have restricted intellectual ability to predict appropriate malnutrition status of a child. In the meantime, human intelligence has great worth for discovering a new intelligent finding. At the same time, artificial intelligence can iterate considerable data but that may have a lack of logical abilities. So far, determining the malnutrition status using deep learning mechanism is one of the most powerful scientific ways to deal with. Especially in a developing nation like Bangladesh where a large number of children are suffering from it, this approach may reduce the number of being malnourished. In our exploration, three classes have been trained separately with respect to the data and applied the model based on individual class on the trained data. But it is also possible to merge all the three class as a single class using Multiclass-Multilabel Classification. It is hoped that, if Multiclass-Multilabel Classification technique is applied then the result will be more accurate and accuracy may be increased. The future study will be deeper and something new knowledge may be discovered which is our new concern as a future scope in the field of artificial intelligence. Furthermore, to fulfill the sustainable development goals (SDGs), maternal and child health is the topmost priority in Bangladesh. Therefore, policymakers and health care providers can easily get the benefit from the deep learning approach to predict the malnourished child beforehand.

REFERENCES

- [1] "WHO. 1995| Physical status: the use and interpretation of anthropometry", Who.int.
- [2] Rahman, Azizur & Biswas, Soma. (2009). Nutritional status of under-5 children in Bangladesh. South Asian Journal of Population and Health. 2. 1-11.
- [3] S. Das and J. Gulshan, "Different forms of malnutrition among under five children in Bangladesh: a cross sectional study on prevalence and determinants", BMC Nutrition, vol. 3, no. 1, 2017.
- [4] "Nutrition country profiles: Bangladesh summary", Fao.org, 2010. [Online]. Available: http://www.fao.org/ag/agn/nutrition/bgd_en.stm.
- [5] A. El-Ghannam, "The Global Problems of Child Malnutrition and Mortality in Different World Regions", Journal of Health & Social Policy, vol. 16, no. 4, pp. 1-26, 2003.
- [6] D. Pelletier and E. Frongillo, "Changes in Child Survival Are Strongly Associated with Changes in Malnutrition in Developing Countries", The Journal of Nutrition, vol. 133, no. 1, pp. 107-119, 2003.
- [7] A. K. Das, A. Ashrafi and M. Ahmmad, "Joint Cognition of Both Human and Machine for Predicting Criminal Punishment in Judicial System," 2019 4th International Conference on Computer and Communication Systems (ICCCS), Singapore, 2019.
- [8] A. K. Das, T. Adhikary, M. A. Razzaque and C. S. Hong, "An intelligent approach for virtual machine and QoS provisioning in cloud computing," The International Conference on Information Networking 2013 (ICOIN), Bangkok, 2013, pp. 462-467.
- [9] A. Tashnim, S. Nowshin, F. Akter and A. K. Das, "Interactive interface design for learning numeracy and calculation for children with autism," 2017 9th International Conference on Information Technology and Electrical Engineering (ICITEE), Phuket, 2017, pp. 1-6.
- [10] A. K. Das, T. Adhikary, M. A. Razzaque, M. Alrubaihan, M. M. Hassan, Z. Uddin, and B. Song, "Big media healthcare data processing in cloud: a collaborative resource management perspective," Cluster Computing, Volume 20, Issue 2, pp 1599-1614, June 2017.
- [11] T. Adhikary, A. K. Das, M. A. Razzaque, M. E. H. Chowdhury and S. Parvin, "Test implementation of a sensor device for measuring soil macronutrients," 2015 International Conference on Networking Systems and Security (NSysS), Dhaka, 2015, pp. 1-8.
- [12] M. Akter, F. T. Zohra and A. K. Das, "Q-MAC: QoS and mobility aware optimal resource allocation for dynamic application offloading in mobile cloud computing," 2017 International Conference on Electrical, Computer and Communication Engineering (ECCE), Cox's Bazar, 2017, pp. 803-808.
- [13] L. Abera, T. Dejene, and T. Laelago, "Prevalence of malnutrition and associated factors in children aged 6–59 months among rural dwellers of damot gale district, south Ethiopia: community based cross sectional study", International Journal for Equity in Health, vol. 16, no. 1, 2017.
- [14] N. Kandala, T. Madungu, J. Emina, K. Nzita and F. Cappuccio, "Malnutrition among children under the age of five in the Democratic Republic of Congo (DRC): does geographic location matter?", BMC Public Health, vol. 11, no. 1, 2011.
- [15] M. Arimond and M. Ruel, "Dietary Diversity Is Associated with Child Nutritional Status: Evidence from 11 Demographic and Health Surveys", The Journal of Nutrition, vol. 134, no. 10, pp. 2579-2585, 2004.
- [16] S. Das and R. Rahman, "Application of ordinal logistic regression analysis in determining risk factors of child malnutrition in Bangladesh", Nutrition Journal, vol. 10, no. 1, 2011.
- [17] K. Bhowmik and S. Das, "On exploring and ranking risk factors of child malnutrition in Bangladesh using multiple classification analysis", BMC Nutrition, vol. 3, no. 1, 2017.
- [18] Vijayakumar S, Deepika MG, Amalendu J, Deepa G. Factors Affecting Infant Mortality Rate in India: An Analysis of Indian States 2016.
- [19] Z. Markos, F. Doyure, M. Yifiru and J. Haidar, "Predicting Under Nutrition Status of Under-Five Children Using Data Mining Techniques: The Case of 2011 Ethiopian Demographic and Health Survey", Journal of Health & Medical Informatics, vol. 5, no. 2, 2014.
- [20] B. Abuya, J. Ciera and E. Kimani-Murage, "Effect of mother's education on child's nutritional status in the slums of Nairobi", BMC Pediatrics, vol. 12, no. 1, 2012.